

LAB 4

TECH 60 - Sec 14

F26

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Objective:

The purpose for Lab 4 was to understand and utilize the Voltage Divider Rule, also known as VDR. Various resistors ranging from 1k Ω all the way to 100 k Ω were connected in series on a breadboard. The 1 k Ω and 2 k Ω were used as R_1 and R_2, and the others acted as the load resistors. The calculations were done by hand, using theoretical, multisim, and measured values. The results did show some discrepancies, especially when comparing the multisim and breadboard values, with the highest percent error being 10.07%.

Equipment/Material:

- Multisim software running on PC
- Various resistors: 1 k Ω (Quantity 2), 2 k Ω , 3 k Ω , 4 k Ω , 5.1 k Ω , 10 k Ω , 100 k Ω .
- Digital Multimeter (DMM)
- Power Supply
- 4 Minigrabber wires

Data:

Part I (Voltage Divider Rule)

hand calculations:

Assuming $V_s = 9$, and $V_2 = 6$

$$V_{R1} = (R_1 / (R_1 + R_2)) V_s$$

$$V_{R1} = (1k\Omega / (1k\Omega + 2k\Omega)) 9V$$

$$V_{R1} = (1/3) 9V$$

$$V_{R1} = 3V$$

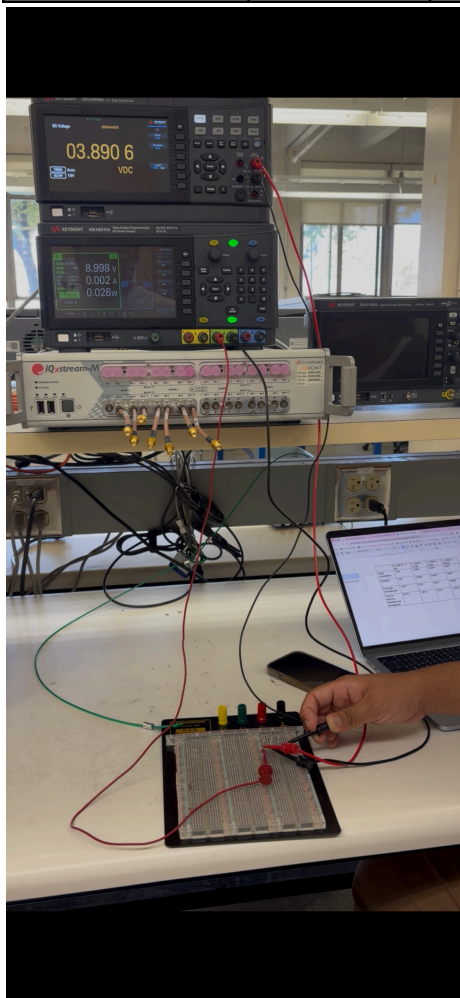
$$V_{R2} = (R_2 / (R_1 + R_2)) V_s$$

$$V_{R2} = (2k\Omega / (1k\Omega + 2k\Omega)) 9V$$

$$V_{R2} = (2/3) 9V$$

$$V_{R2} = 6V$$

	V_s	V_{R1}	V_{R2}
Hand Calculations	9V	3V	6V
Multisim	9V	3V	6V
Prototype (Breadboard)	8.991V	2.995V	5.99V
%Error Between Multisim and Breadboard	0.1%	0.17%	0.17%





Part II (Voltage Divider Rule Loading Effects)

Hand Calculations:

Assuming $V_s = 9V$, $V_{R1} = 3V$, $V_{R2} = 6V$

$$V_L = V_2 = \left[\frac{R_2 \parallel R_L}{R_1 + (R_2 \parallel R_L)} \right] V_s$$

$R_L = 1 \text{ k}\Omega$:

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L)$$

$$V_L = [((R_2 * R_L) / (R_2 + R_L)) / (R_1 + (R_2 * R_L) / (R_2 + R_L))] * V_s$$

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L) = (2 * 1) / (2 + 1) = 0.667$$

$$V_L = [(R_2 \parallel R_L) / (R_1 + (R_2 \parallel R_L))] V_s$$

$$V_L = [0.667 / (1 + 0.667)] * 9 = 3.601 = 3.6V$$

$R_L = 5.1 \text{ k}\Omega$:

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L)$$

$$V_L = [((R_2 * R_L) / (R_2 + R_L)) / (R_1 + (R_2 * R_L) / (R_2 + R_L))] * V_s$$

$$V_L = [((2 \text{ k}\Omega * 5.1 \text{ k}\Omega) / (2 \text{ k}\Omega + 5.1 \text{ k}\Omega)) / (1 \text{ k}\Omega + (2 \text{ k}\Omega * 5.1 \text{ k}\Omega) / (2 \text{ k}\Omega + 5.1 \text{ k}\Omega))] * 9V$$

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L) = (2 * 5.1) / (2 + 5.1) = 1.437$$

$$V_L = [(R_2 \parallel R_L) / (R_1 + (R_2 \parallel R_L))] V_s$$

$$V_L = [1.437 / (1 + 1.437)] * 9 = 5.307V$$

$R_L = 10 \text{ k}\Omega$:

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L)$$

$$V_L = [((R_2 * R_L) / (R_2 + R_L)) / (R_1 + (R_2 * R_L) / (R_2 + R_L))] * V_s$$

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L) = (2 * 10) / (2 + 10) = 1.667$$

$$V_L = [(R_2 \parallel R_L) / (R_1 + (R_2 \parallel R_L))] V_s$$

$$V_L = [1.667 / (1 + 1.667)] * 9 = 5.625V$$

$$R_L = 100 \text{ k}\Omega$$

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L)$$

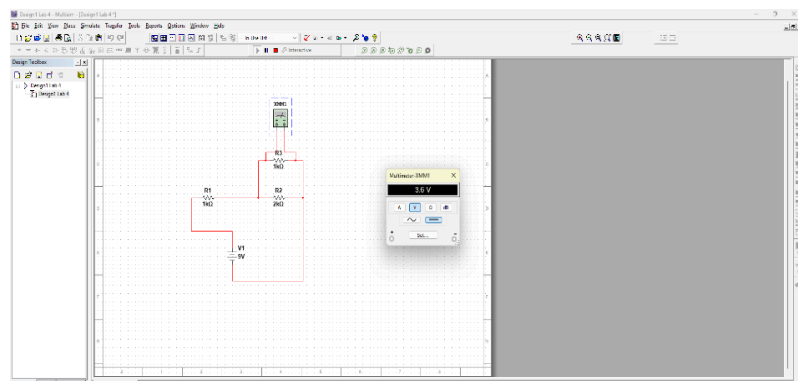
$$V_L = [((R_2 * R_L) / (R_2 + R_L)) / (R_1 + (R_2 * R_L) / (R_2 + R_L))] * V_s$$

$$R_2 \parallel R_L = R_2 * R_L / (R_2 + R_L) = (2 * 100) / (2 + 100) = 1.961$$

$$V_L = [(R_2 \parallel R_L) / (R_1 + (R_2 \parallel R_L))] V_s$$

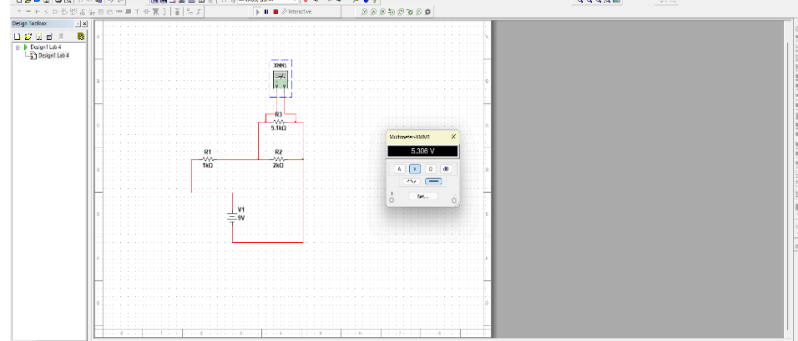
$$V_L = [1.961 / (1 + 1.961)] * 9 = 5.96V$$

	V_L with R_L = 1 kΩ	V_L with R_L = 5.1 kΩ	V_L with R_L = 10 kΩ	V_L with R_L = 100 kΩ
Hand Calculations	3.6V	5.307V	5.625V	5.96V
Multisim	3.6V	5.306V	5.625V	5.96V
Prototype (Breadboard)	3.89V	4.99V	5.18V	5.36V
%Error Between Multisim and Breadboard	8.06%	5.96%	7.91%	10.07%



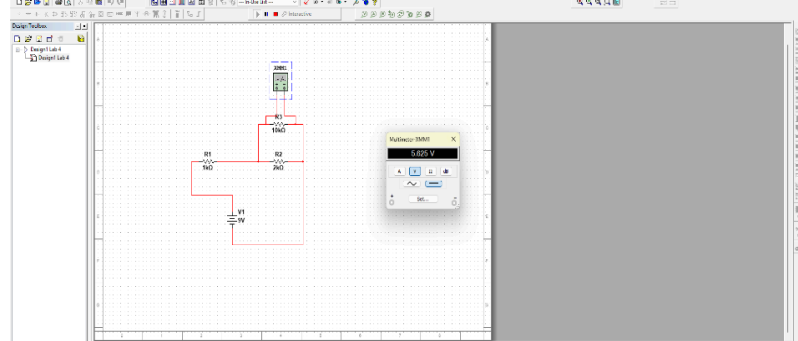
-----Checking SPICE model for Design Lab 1 - Analogous, Subcircuit: V1, R1, R2, V1, 0.000 s-----
Warning: No ground node was found in your circuit. To ensure accurate simulation, add a ground component and simulate again.
***** SPICE Model check completed. (Error): Terminating *****

Results | View | Comments | Copy Items | Simulation
Instrument: RefDes(MAN), Hierarchical: Location(B)
Design Lab 1 - Analogous (Design Lab 1)
File Edit View Design Simulation Tools Reports Options Windows Help
-----in the lab-----
Design-Tree
Design Lab 1
Design Lab 1



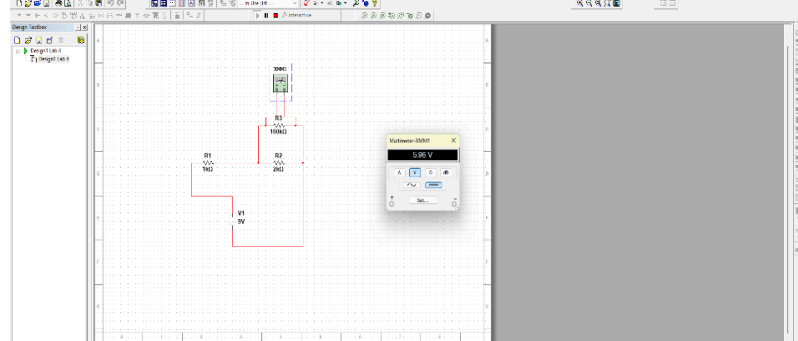
-----Checking SPICE model for Design Lab 4 - Analogous, Subcircuit: V1, R1, R2, V1, 0.000 s-----
Warning: No ground node was found in your circuit. To ensure accurate simulation, add a ground component and simulate again.
***** SPICE Model check completed. (Error): Terminating *****

Results | View | Comments | Copy Items | Simulation
Instrument: RefDes(MAN), Hierarchical: Location(B)
Design Lab 4 - Analogous (Design Lab 4)
File Edit View Design Simulation Tools Reports Options Windows Help
-----in the lab-----
Design-Tree
Design Lab 4
Design Lab 4



-----Checking SPICE model for Design Lab 4 - Analogous, Subcircuit: V1, R1, R2, V1, 0.000 s-----
Warning: No ground node was found in your circuit. To ensure accurate simulation, add a ground component and simulate again.
***** SPICE Model check completed. (Error): Terminating *****

Results | View | Comments | Copy Items | Simulation
Instrument: RefDes(MAN), Hierarchical: Location(B)
Design Lab 4 - Analogous (Design Lab 4)
File Edit View Design Simulation Tools Reports Options Windows Help
-----in the lab-----
Design-Tree
Design Lab 4
Design Lab 4



-----Checking SPICE model for Design Lab 4 - Analogous, Subcircuit: V1, R1, R2, V1, 0.000 s-----
Warning: No ground node was found in your circuit. To ensure accurate simulation, add a ground component and simulate again.
***** SPICE Model check completed. (Error): Terminating *****

Results | View | Comments | Copy Items | Simulation
Instrument: RefDes(MAN), Hierarchical: Location(B)
Design Lab 4 - Analogous (Design Lab 4)
File Edit View Design Simulation Tools Reports Options Windows Help
-----in the lab-----
Design-Tree
Design Lab 4
Design Lab 4

Questions:

1. What is the effect of load on the load voltage of a voltage divider.

Adding a load resistor pulls the load voltage down, and because they are in parallel with R2, they produce a combined resistance smaller than either resistor alone. This lowers the voltage.

2. How do you minimize the loading effects in a voltage divider.

You make sure that the RL is much larger than R2 so that the loaded voltage stays close to the unloaded value.

3. Compare the measured source voltage to the hand calculation values of this quantity.

Should they be the same value ideally? Explain briefly why they are not exactly equal.

Yes, they should be the same ideally. However, the hand calculations and the Multisim do not account for the human error and environmental factors that play within a real circuit.

4. Compare the measured resistor voltage values to the hand-calculated values. Explain briefly why they are not exactly equal.

The actual resistor values in practice are not exactly 1 k Ω and 2 k Ω , so the resistors are slightly lower in resistance than their marketed values. Some factors include the contact resistance as well as the human error associated with holding and connecting the components. Also, the DMM measurement accuracy could be a factor in this reading.

5. From your experimental results and hand calculated values have you confirmed the voltage divider rule?

Yes, we have confirmed the voltage divider rule by comparing our real-world values to the theoretical hand calculations that were completed beforehand.

Challenges: We had to face hand calculations because there were some misconceptions when calculating the resistors but somehow we managed to fix that issue. Another challenge we had faced was trying to measure the resistors because the machines were reading the resistors well. We had to reset the meters multiple times to get the meters to work and it ended up working.

Discussion/Conclusions: We learned on how to work on the voltage divider rule and how loading affects the output voltage. Our hand calculations and multisims results were very similar, which confirmed our calculations. We found that smaller load resistors caused a larger decrease in voltage, while larger resistors kept the voltage closer to the unloaded value. Overall, the lab helped us better understand voltage dividers and loading effects.